

Problem Set 5

Submission:

Friday, 27/03/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.

Only problems marked with (*) will be graded.

1. (*) σ -algebras and random variables

[4 Points]

In a random experiment, a green and a blue die are tossed at the same time. We choose $\Omega = \{1, \dots, 6\}^2$ (with (ω_1, ω_2) denoting the numbers shown by the (blue, green) die) and consider the following two σ -algebras:

$$\mathcal{F} = \mathcal{P}(\Omega),$$

$$\mathcal{F}_{\text{sym}} = \{A \subseteq \Omega : \text{for all } (\omega_1, \omega_2) \in \Omega, (\omega_1, \omega_2) \in A \Leftrightarrow (\omega_2, \omega_1) \in A\}.$$

- (a) Note that $\mathcal{F}_{\text{sym}}$ contains only sets that are symmetric with respect to the exchange of the order of ω_1 and ω_2 . In which situations does it make sense to use $\mathcal{F}_{\text{sym}}$ to model a probability space? Give a concrete example.
- (b) Show that $\mathcal{F}_{\text{sym}}$ is indeed a σ -algebra.
- (c) Consider the two subsets of Ω :

$$A = \text{"one die shows the number 3"},$$

$$B = \text{"the blue die shows the number 3"}.$$

Show that $A \in \mathcal{F}_{\text{sym}}$ but $B \notin \mathcal{F}_{\text{sym}}$.

- (d) Consider the two maps

$$X : \begin{cases} \Omega \rightarrow \mathbb{R} \\ (\omega_1, \omega_2) \mapsto \omega_1 \end{cases} \quad \text{"number shown by the blue die"}$$

$$S : \begin{cases} \Omega \rightarrow \mathbb{R}, \\ (\omega_1, \omega_2) \mapsto \omega_1 + \omega_2 \end{cases} \quad \text{"sum of numbers shown by both dice"}$$

Which of these is/are a random variable/random variables on $(\Omega, \mathcal{F}_{\text{sym}})$?

2. Calculations with probability density functions

Consider spherical bubbles of radius R that form randomly in an aerated aquarium. Assume first that the radius (measured in mm) of a bubble follows a uniform distribution $R \sim \mathcal{U}([1, 10])$.

- (a) Calculate the probability density function of the random volume of a bubble.

For the rest of the assignment, we assume that the radius follows a *lognormal* distribution with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, $R \sim \ell\mathcal{N}(\mu, \sigma^2)$, which by definition fulfills

$$R \stackrel{d}{=} \exp(Y), \quad Y \sim \mathcal{N}(\mu, \sigma^2).$$

- (b) Find the probability density function of the law of R , that is find the probability density function of $\ell\mathcal{N}(\mu, \sigma^2)$.

- (c) Show that if R is lognormal distributed with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, then the volume V and the surface area A of the bubble are also lognormal-distributed, and find the corresponding parameters in terms of μ and σ .

3. Cumulative distribution functions

Let X be a real random variable with cumulative distribution function

$$F_X(x) = \begin{cases} \frac{1}{4}e^{x+2} + c, & x < -2, \\ \frac{1}{2} + \frac{x}{8}, & -2 \leq x < 0, \\ \frac{2+x}{3+x}, & 0 \leq x < 5, \\ (\lambda - 1)e^x + 1, & x \geq 5. \end{cases}$$

- (a) Determine the values of the real constants c, λ .
 (b) Calculate $\mathbf{P}[0 \leq X < 5]$ and $\mathbf{P}[-2 < X \leq 5]$.

Calculate the cumulative distribution function for random variables with the following distributions and draw a sketch for each:

- (c) $X \sim \text{Bin}(4, \frac{1}{4})$,
 (d) $Y \sim \text{Cau}(1, 2)$, where $\text{Cau}(\mu, a)$ stands for the Cauchy distribution with parameters $\mu \in \mathbb{R}$ and $a > 0$, characterized by the density

$$f_Y(y) = \frac{a}{\pi(a^2 + (y - \mu)^2)}, \quad y \in \mathbb{R}.$$

Hint: From Calculus, it is known that $\arctan'(x) = \frac{1}{1+x^2}$, where $\arctan$ (sometimes written as $\tan^{-1}$) is the inverse of the tangent function $\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$.

4. (*) Medians of distribution functions

[4 Points]

A real number m is called a *median* of the distribution function F whenever

$$(*) \quad \lim_{y \uparrow m} F(y) \leq \frac{1}{2} \leq F(m).$$

- (a) Show that $(*)$ is equivalent to the requirement $\mathbf{P}[[m, \infty)) \geq \frac{1}{2}$ and $\mathbf{P}[(-\infty, m]] \geq \frac{1}{2}$ for the probability measure $\mathbf{P}$ associated with F .
 (b) Show that every distribution function F has at least one median.
Hint: Consider $\alpha = \sup\{x \in \mathbb{R} : F(x) < \frac{1}{2}\}$, show that α satisfies $(*)$.
 (c) Show that the set of medians of F is a closed interval of $\mathbb{R}$.
Hint: Consider $\beta = \sup\{x \in \mathbb{R} : F(x) \leq \frac{1}{2}\}$, show that the median fulfills $m \in [\alpha, \beta]$.
 (d) Show that $F(m) = \frac{1}{2}$ for any median m if F is continuous.